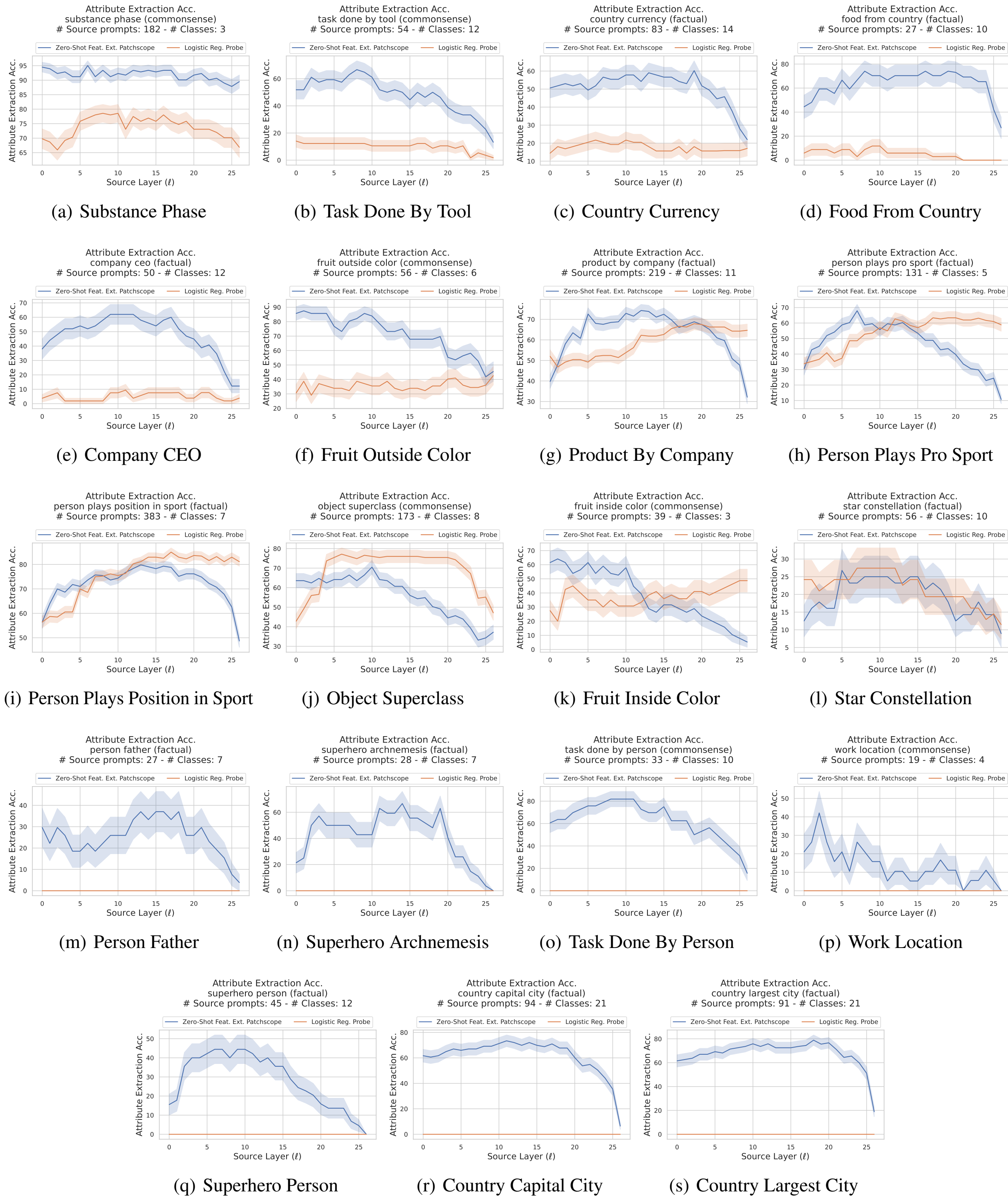


(a) Substance Phase (b) Task Done By Tool (c) Country Currency (d) Food From Country

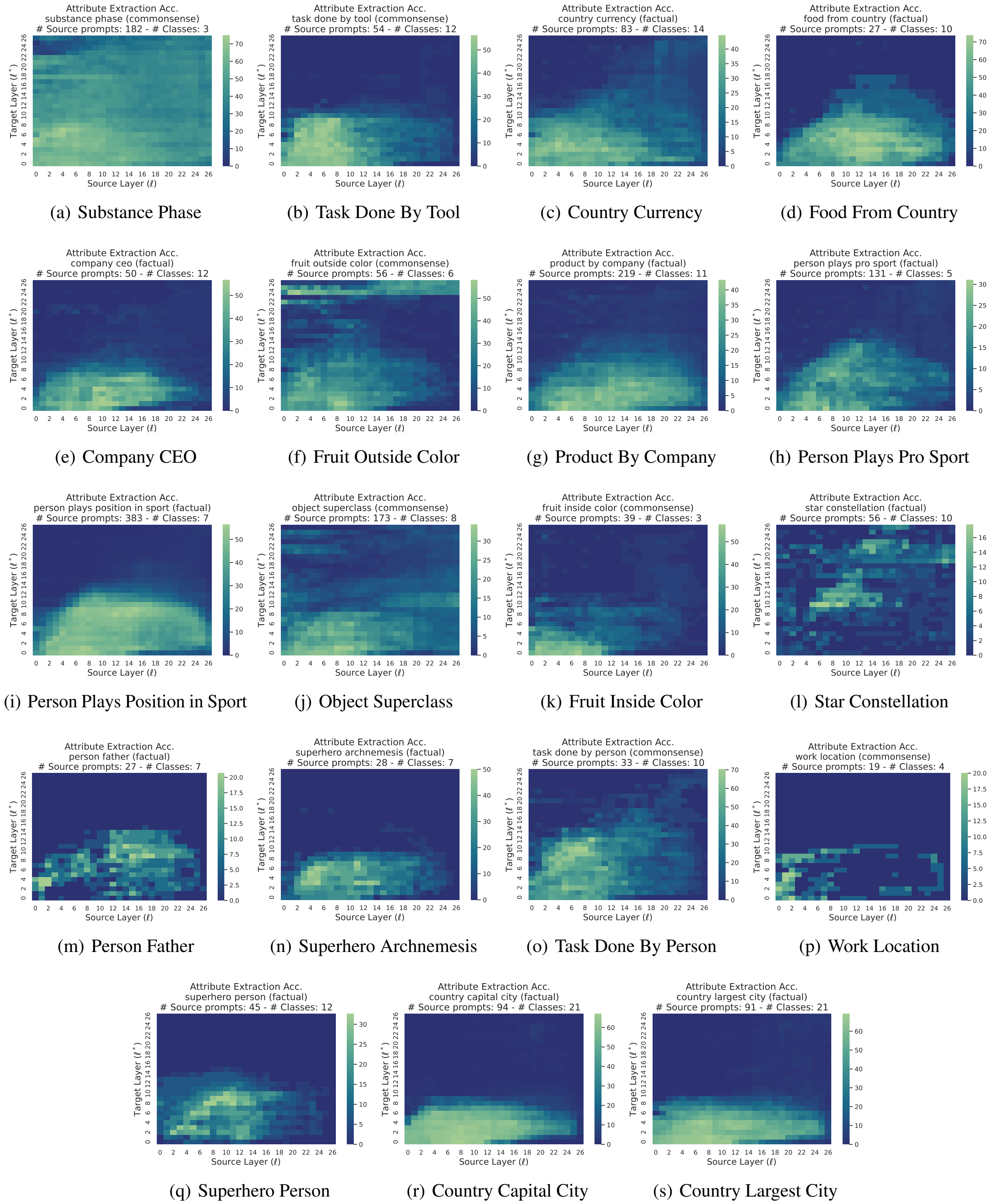
(e) Company CEO (f) Fruit Outside Color (g) Product By Company (h) Person Plays Pro Sport

(i) Person Plays Position in Sport (j) Object Superclass (k) Fruit Inside Color (l) Star Constellation

(m) Person Father (n) Superhero Archnemesis (o) Task Done By Person (p) Work Location

(q) Superhero Person (r) Country Capital City (s) Country Largest City

Figure 6. Feature extraction accuracy with respect to source layer (ℓ) across various factual and commonsense reasoning tasks. The zero-shot feature extraction `Patchscope` works consistently better than the logistic regression probe in early layers, and mostly in mid layers. There is a decline in `Patchscope` accuracy in later ℓ as the source representations shift toward next-token prediction.



(a) Substance Phase (b) Task Done By Tool (c) Country Currency (d) Food From Country

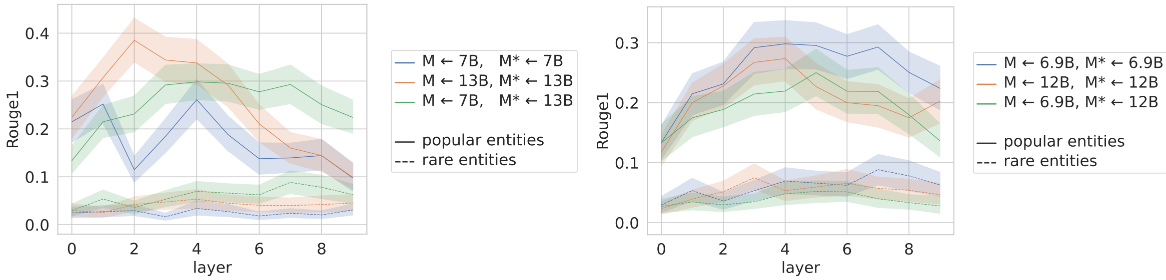
(e) Company CEO (f) Fruit Outside Color (g) Product By Company (h) Person Plays Pro Sport

(i) Person Plays Position in Sport (j) Object Superclass (k) Fruit Inside Color (l) Star Constellation

(m) Person Father (n) Superhero Archnemesis (o) Task Done By Person (p) Work Location

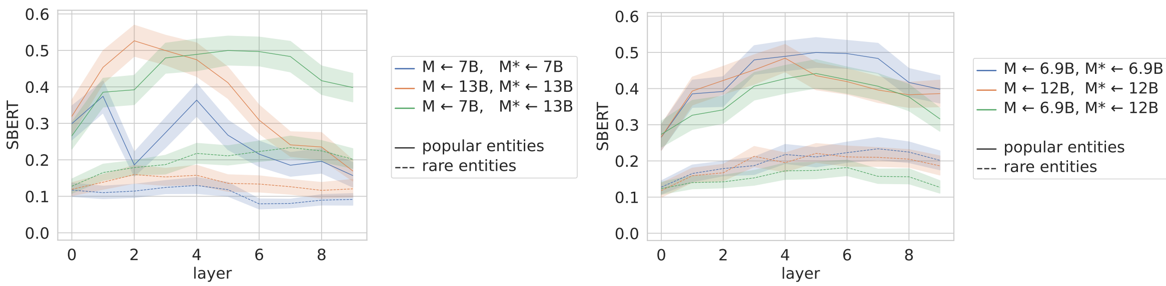
(q) Superhero Person (r) Country Capital City (s) Country Largest City

Figure 7. The interaction between source and target layers in the zero-shot feature extraction Patchscope across various factual and commonsense reasoning tasks. Each cell (ℓ, ℓ^*) in the heatmap shows the attribute extraction success rate where source and target layers are fixed to ℓ and ℓ^*, respectively. Particularly, there is a higher success rate in the lower left quadrants, representing early to mid source and target layer combinations. The right half of the heatmaps shows late source layers, where the source representation has shifted toward next-token prediction, leading to a lower success rate in attribute extraction. The top half of the heatmaps shows late target layers. When the accuracy is a function of more than a single next-token, the placeholder token representation still remains in the early layers, leading to a lower attribute extraction rate.



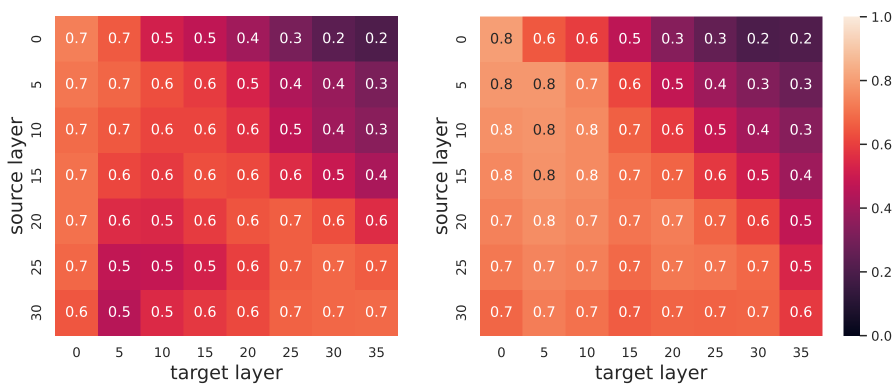
(a) Vicuna (b) Pythia

Figure 8. Rouge1 scores of the generated descriptions against descriptions from Wikipedia.



(a) Vicuna (b) Pythia

Figure 9. SBERT scores of the generated descriptions against descriptions from Wikipedia.



(a) Vicuna: $\mathcal{M} \leftarrow$ 7B, $\mathcal{M}^* \leftarrow$ 13B (b) Pythia: $\mathcal{M} \leftarrow$ 6.9B, $\mathcal{M}^* \leftarrow$ 12B

Figure 10. Precision@1 scores (↑ is better) on next-token prediction estimation in Vicuna and Pythia with cross-model `Patchscopes`

as Vicuna 13B processes "`Will Smith`", it goes from *"Smithsonian Museum"* to the *"Smith rock band"* to the actor and rapper, *"Will Smith"*. However, Pythia 12B starts with *"Smith & Wesson weapon manufacturing company"* before it resolves the entity as the American actor, *"Will Smith"*.

We also observe another phenomenon which we refer to as placeholder contamination. It is the case where the remaining representation of the placeholder entity "`x`" in the early layers interferes with the model's capability in generating descriptions for the patched token. For example, see Vicuna 13B response to "`Paris Hilton`" entity in Tab. 6. First, we see the gradual process of going from *"Rigatoni"* pasta, to *"Hilton Hotels"*, to the socialite *"Paris Hilton"* in layers 1-6. But in layers 7 and onward, the generation seems to describe the placeholder token "`x`"rather than the "`Paris Hilton`" entity: *"Placeholder for a variable or concept"*, *"Variable representing any number of things or concepts"* or *"x is a placeholder"*. For future, we would like to quantify these qualitative observations, study to what extend this contamination can be mitigated with a different placeholder choice, and why some models might be more susceptible to this contamination than others.